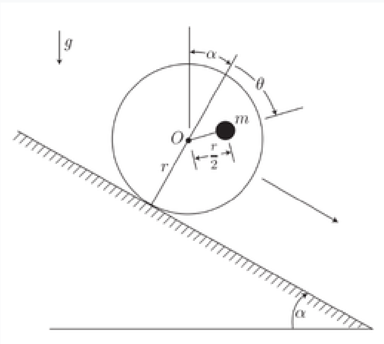
Problem 1

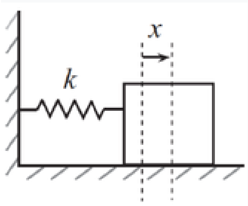
Problem 1

A disk of radius r and negligible mass (i.e., assume the disk is massless) has a particle of mass m embedded at a distance $\frac{1}{2}r$ from the center O, as shown in the figure. Assume the disk rolls without slipping down a plane inclined at an angle α from the horizontal. Use θ as the single degree of freedom for this system. Derive the Hamilton's equations of motion for θ, p_θ.



Problem 2

Problem 2



When we use Hamiltonians, we can define canonical coordinate changes (under certain conditions) from (q, p) to (Q, P) by using *generating functions* $F(q, Q, t)$. With this function in terms of the old and new coordinates, the conjugate momenta are defined by,

$$p = \frac{\partial F}{\partial q}, \quad P = -\frac{\partial F}{\partial Q}.$$

Using these relationships, we can write a new Hamiltonian $H'(Q, P)$, which follows the same canonical equations of motion.

$$\dot{Q} = \frac{\partial H'}{\partial P}, \quad \dot{P} = -\frac{\partial H'}{\partial Q}$$

As an example, consider a simple harmonic oscillator with mass $m = 1$ and a spring coefficient k. Let $\omega = \sqrt{k}$. Let $q = x$ be our only generalized coordinate.

(a) Derive the Hamiltonian for this system in terms of q, p, ω in the usual way. You should find that,

$$H(q, p) = \frac{1}{2}\left(p^2 + \omega^2 q^2\right).$$

(b) Choose the following generating function (I know it seems ridiculous, trust me):

$$F(q, Q) = \frac{1}{2}\omega q^2 cot(2\pi Q)$$

(c) Use the relationships above to derive $p(q, Q)$ and $P(q, Q)$.

(d) From the expression for P, you solve for $q(Q, P)$, and then substitute this into the expression for p to find $p(Q, P)$.

(e) Now find $H'(Q, P) = H(q(Q, P), p(Q, P)$. ($H'$ is a new Hamiltonian in terms of Q and P, not a derivative.)

(f) Write the equations of motion of Q, P. You will find that $\dot{Q}$ =constant and $\dot{P} = 0$!

(g) Integrate to find $Q(t), P(t)$ and plug into $q(Q, P), p(Q, P)$. You will have found functions of time (in terms of the constant P)! (One textbook author calls this cracking a peanut with a sledgehammer.")

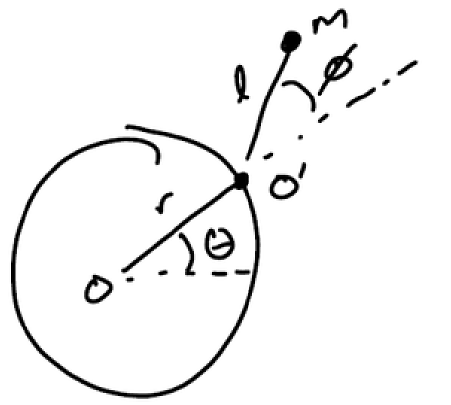
There are three other types of generating functions ($F(q, P), F(p, Q), F(p, P)$) that can be used to canonically change coordinates under certain conditions. We won't fully go into these approaches in this class, but I wanted to show a simple example.

Problem 3

Problem 3

A particle of mass m is attached by a string of length l to a point O' which moves around a hoop of radius r at a constant angular rate $\dot{\theta} = \omega_0$. The angle between the radial direction and the string is given by ϕ. This setup is moving horizontally on a table top (i.e., gravity points into the page).

Apply Hamilton's Equations of Motion for the generalized coordinate $q = \phi$ and its conjugate momentum p.



Problem 4

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Appendix: Code Listings

Problem 1 Code

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Listing 1: TODO: caption once Problem 1's solution/code is written.